

Syllabus

PHYS777 – BOOTSTRAP METHODS IN THEORETICAL PHYSICS I 2026 – 2027 FALL SEMESTER

General

(1) This syllabus has been prepared based on *Middle East Technical University Rules and Regulations Governing Graduate Studies* and applies only to the course and the semester in the subtitle.¹ For any questions, feel free to contact me.²

(2) If you have official documentation of a disability, you may apply for a *Letter for Academic Accommodations* by contacting *METU Disability Support Office*: please do so if this applies. Even if you do not have an officially documented disability, contact me if you have any special situation such as *physical disability, learning disability, ADHD, chronic health issue, mental health condition, complications with a pregnancy, temporarily disabling injury or circumstance, upcoming surgery* and so on.³

(3) The METU Code of Honor and the academic integrity guide, available via the *METU Code of Ethics & Core Values* website, are applicable to this course. For further details, please refer to https://soneralbayrak.com/teaching/academic_integrity.pdf.

(4) My weekly schedule, my office hours, possibility of appointments, and any other related information are available at <https://soneralbayrak.com/schedule>.

Prerequisites

(5) Both Phys777 and Phys778 are research-level elective courses designed for graduate students in the Department of Physics. Although the courses are rather self-contained, undergraduate-level physics knowledge is a prerequisite to follow them properly; that being said, any ambitious undergraduate student with a sufficient command of undergraduate physics is welcome to take these courses as well.

Course Timeline and Attendance

(6) The weekly lecture schedule:⁴

Lecture-1	Thursday	10:40 – 11:45	@ P421
Break		11:45 – 11:55	
Lecture-2	Thursday	11:55 – 13:00	@ P421

Note that the course times are not standard, but this ensures that students have at least half an hour for lunch before the next classes start at 13:40.

¹Any topic that is not explicitly discussed in this document, such as the medium of instruction or the available letter grades, is governed by the *Middle East Technical University Rules and Regulations Governing Graduate Studies*.

²My official email: soneralb@metu.edu.tr.

³You should contact me as soon as a need for an academic accommodation arises so that I can do my best to tailor your learning experience to your needs: please do not delay or postpone informing me of your situation!

⁴See <https://www.metu.edu.tr/location-classrooms-metu-campus> for the location of classrooms.

(7) Attendance is mandatory by the regulations; however, it will not affect your grade in this course.

Course Content

Overview

(8) For the official details on course objectives, course goals, and course learning outcomes, see the METU Catalog webpage:

https://catalog.metu.edu.tr/course.php?course_code=2300777

(9) More informally, this course series introduces several interconnected concepts in theoretical physics. Although there is a sizable overlap with standard field theory courses (such as quantum field theory), bootstrap methods are inherently non-perturbative, and the focus is on the implications of the conceptual points rather than on the actual computations (unlike a field theory course).

(10) As research level courses, neither Phys777 nor Phys778 can be fully followed through a book; nevertheless, many of the essential ingredients can be found in the following books:

- M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Cambridge University Press, 2014.
- H. Elvang and Y.-t. Huang, *Scattering Amplitudes in Gauge Theory and Gravity*, Cambridge University Press, 2015.
- V. K. Dobrev, G. Mack, V. B. Petkova, S. G. Petrova, and I. T. Todorov, *Harmonic Analysis on the n -Dimensional Lorentz Group and Its Application to Conformal Quantum Field Theory*, Lecture Notes in Physics 63, Springer, 1977.
- E. S. Fradkin and M. Ya. Palchik, *Conformal Quantum Field Theory in D-Dimensions*, Mathematics and Its Applications, Kluwer Academic Publishers, 1996.

(11) Similar courses have been taught worldwide, and some of them have freely available lecture notes online, including but not limited to

- Lecture notes of Slava Rychkov at various schools (EPFL, TASI, CEA Saclay, ...)
- Lecture notes of David Simmons-Duffin at various schools (Caltech, TASI, ...)
- Lecture notes of Marc Gillioz at SISSA
- Lecture notes of Shai Chester at Weizmann Institute
- Lecture notes of Joshua Qualls at National Taiwan University
- Lecture notes of Bert Schellekens at Saalburg
- Lecture notes of Paul Ginsparg at Les Houches
- Lecture notes of Lorenz Eberhardt at DESY
- Lecture notes of Yu Nakayama at Taiwan Central University
- Lecture notes of V. S. Dotsenko at Kyoto University

Phys777

(12) **Concept of critical phenomena:** Review of phase diagrams, ensembles, and free energy; Phenomenological approach to critical points, Landau potential, spontaneous symmetry breaking

(13) **Euclidean (Wiener) Path Integrals:** Gaussian path integrals, generating functions, cumulant expansion, Wick's contraction; Correlation functions, correlation length; Critical exponents, universality

(14) Renormalization Group Flow: Position space RG, momentum space RG; Classification of fixed points; Analysis of the Wilson–Fisher fixed point; Epsilon-expansion; Scaling of operators, relevance under RG flow, scale invariance

(15) Conformal symmetry: Conformal symmetry as the natural extension of scale invariance; Dilatation and conformal groups; Irreducible representations of the conformal group; Classification of operators under the action of the conformal group

(16) Symmetry of Maxwell Theory: Stress-energy tensor, virial current; Conformal (scale) symmetry of Maxwell Theory in $d = 4$ ($d \neq 4$)

(17) Introduction to Conformal Bootstrap: Operator Product Expansion (OPE), Associativity of OPE, Crossing Symmetry; Conformal kinematics, conformal frame; Casimirs of the conformal group, conformal blocks, hypergeometric functions; Unitarity bounds, lightcone limit, spectrum of CFTs; Numerical tools in the conformal bootstrap

Phys778

(18) Mathematical Review of Special Relativity: Lorentz group, its subgroups, and associated algebras; Relation between the Lorentz and rotation groups: imaginary time and Wick rotation; Stabilizer subgroup formalism (Wigner classification); Representations of the Lorentz group, spin; Lorentz covariance of physical observables

(19) Axiomatic Approach to Quantum Mechanics: Hilbert space and its properties; Pictures, Møller operators; Euclidean & Lorentzian time evolution; Distributional nature of Lorentzian operators and basic introduction to distributions

(20) Lorentzian (Feynman) Path Integrals: Feynman’s approach to quantum mechanics: quantization via path integrals, Feynman diagrams; Connection between Feynman & Wiener path integrals; Branch cuts of Lorentzian correlators, time ordering

(21) Scattering Matrix: Haag–Ruelle construction, LSZ theorem; Analyticity of the S-matrix, implications of causality; Locality vs Causality vs Cluster decomposition; Generalization to homogeneous spaces: an active area of research

(22) Concept of Unitarity: Optical theorem, cutting rules, spectral decomposition

(23) Bootstrap Philosophy & Modern non-perturbative techniques: Perturbative vs non-perturbative, exact vs constrained results, analytic vs numeric approaches; Global symmetries and how they can be utilized; Coleman–Mandula theorem; On-shell recursion relations, BCFW, spinor helicity, amplitudes program; Positive geometry, cosmological polytopes

Grading, Examinations, and Make-up

(24) The course has a single midterm examination and a single final examination, both conducted as oral presentations.

(25) Students are free to use any assistance they can receive to prepare the presentations:

- they can ask other students, assistants, professors, or even passing strangers for help;
- they can use any printed or online resources;
- they can (and are in fact **encouraged to**) utilize any large language model or any other form of artificial intelligence.

(26) The grading rubric is given in Table 1: analyze it to prepare appropriately for the exams.

Table 1: Grading rubric for the presentations

Main parts	What to grade (each item is worth 10 points)	Suggestions
Sufficiency of presentation material	- The opening was good: it was suitable for the audience, included a hook to capture attention, summarized the talk well, contained sufficient motivation as to why we should be interested. - The talk had a conclusion/summary part that wrapped up all the important points; it included a quick discussion of the current literature and the open directions in the field; and so on	This part is mostly algorithmic and can be heavily optimized with artificial intelligence: utilize it to the fullest extent!
Delivery of the talk	- The talk was professional: the speaker seemed confident and sufficiently knowledgeable about their talk; they spoke at a pace easy to follow, and so on. - The talk was prepared to fit the allocated time, and the speaker delivered it so that it neither ended too early nor was left unfinished when the time ran out. - The student did not simply memorize things but actually understood the content and its physical importance.	This is more art than craft! You <i>have to practice</i> how to talk in front of a crowd, how to seem confident, how to adjust your pace and so on: there is no shortcut.
Handling of the questions	- The student was able to maintain the flow of the talk despite my repeated interruptions through questions. - The student understood my questions, or at least faked it sufficiently convincingly. - The student was able to answer my questions correctly or at least showed an acknowledgement of their lack of sufficient knowledge. - No scientifically or mathematically incorrect information was given at any point during the talk. - I learned something new from the student through my questions.	Understand how I view this whole exam: Cem used AI to prepare the whole presentation, memorized everything, and repeated it in class; Ceren used AI to learn the content; the only way I can tell Ceren apart from Cem is by observing <i>how they answer my questions during their talk</i> .

(27) The semester score of a student is the average of the exam scores, and letter grades are given based on semester scores, with the specifics to be determined at the end of the semester based on the class performance.

(28) If you miss any exam without a medical report, you receive the letter grade NA. If you miss an exam and have a medical report, please contact me immediately so that we can arrange a make-up examination for you.

(29) Exam locations are announced via email in due time.

(30) Exams will take place within the official class hours (10:40 – 13:30) on class days; the details can be seen via the semester calendar in Figure 1.

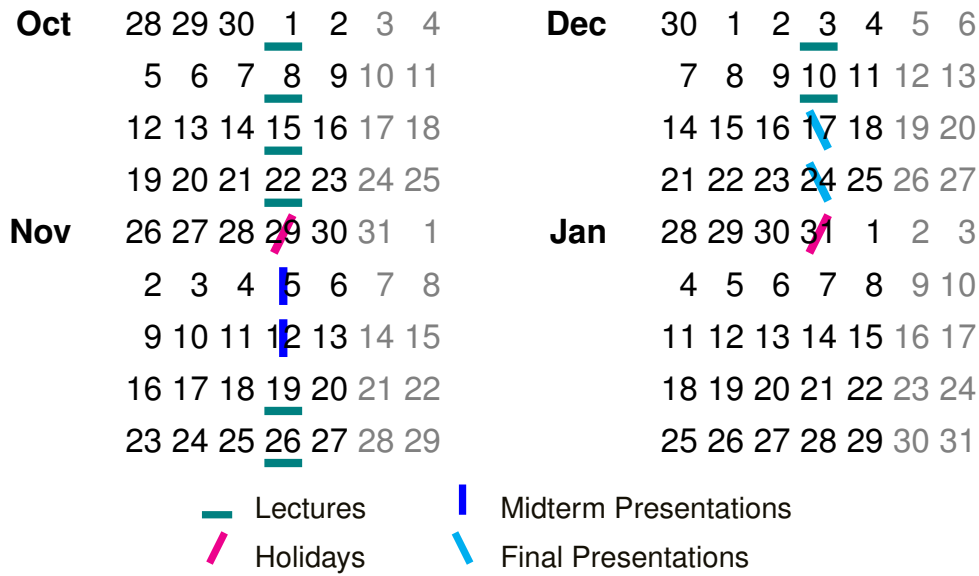
(31) My deliberate choice for the design of the exams can be best understood with the following observations:

- This is a research-level elective course, aimed for students interested in staying inside the academia.
- With the rapid advance in artificial intelligence, especially through large language models (LLM's), specialized knowledge becomes easier to access, meaning that *knowledge is less valuable compared to what it used to be*; therefore, other skills have become far more essential to survive in academia!

Through practising for examinations, students are expected to develop towards following skills:

- conducting comprehensive literature search (ideally benefiting from LLM's as tools);
- preparing lectures and presentations (ideally benefiting from LLM's as tools);
- understanding, however partially, some of the fundamental papers in (or conceptually related to) conformal bootstrap (ideally benefiting from LLM's as tools);
- delivering talks even without full mastery of the content (prepared in limited time).

Figure 1: Semester Calendar



Midterm Examination: *Trial Lecture*

(32) In this oral examination, the student will act as an instructor and give a short lecture aimed at a hypothetical class of advanced-level senior undergraduate physics students.

(33) Within the week following the add-drop period, I will email the duration of the talks, in which classroom the exam will take place, and the topic for each student.

(34) I will take student preferences into consideration if they send me an email (before the end of the add-drop period) containing their three preferred topics among the following:

The Drude model	Brownian motion	Blackbody radiation	Bose–Einstein condensation
Landau theory	The Ising model	Quantum tunneling	The van der Waals fluid
Bloch’s theorem	Quantum eraser	Noether’s theorem	The Aharonov–Bohm effect
Bell’s theorem	Landau levels	The Meissner effect	Relativistic causality
Maxwell’s demon	Gauge invariance	Fermi’s golden rule	Covariant electrodynamics

(35) The presentations should be compiled into a PDF file (under 10 MB) and sent to me via email by the end of October.

(36) A PDF sent late, on November X , will mean the midterm being graded out of $\max\{0, (5 - X) \cdot 20\}$ instead of 100.

Final Examination: *Accelerated Paper Defence*

(37) In this oral examination, the student will act as the author of a chosen paper and give a hypothetical short departmental seminar about that work.

(38) Within the week following the add-drop period, I will email the duration of the talks and in which classroom the exam will take place.

(39) Potential papers to be chosen are clustered into 6 categories and listed in Article (43): a student can choose 2 or 3 clusters and send me their choice (such as $\{A, B, C\}$)⁵ before the end of November via email. I will then choose their paper from that list.

⁵The list should contain at least 15 papers.

(40) I will email each student 72 hours before their final talk and tell them which paper they will be defending; in other words, they will have only 72 hours to prepare for that particular paper, hence the *accelerated* paper defence.

(41) The presentations should be compiled into a PDF file (under 10 MB) and sent to me via email at least 24 hours before their talk, meaning that students will have 48 hours to prepare the presentation PDF.

(42) If a student fails to send the presentation PDF in time, then they will be graded out of 50 instead of 100.

(43) One of the following papers will be chosen for each student:

— Cluster A — Scaling and universality before the renormalization group

Paper	Comment
J. Chem. Phys. 43, 3898	The scaling hypothesis stated as a homogeneity property of the free energy; the exponent relations follow before anyone could explain them. Translation to a general audience is free: it is thermodynamics.
Physics 2, 263	Block spins: the physical picture that Wilson later turned into equations, and the entry point for the correlation length and for coarse-graining.
Phys. Rev. Lett. 17, 1133	Fluctuations destroy continuous order in $d \leq 2$; the lower critical dimension made rigorous.
Phys. Rev. Lett. 32, 292	Landau theory predicts a second-order transition; fluctuations of the gauge field make it first order. Coleman–Weinberg seen from a superconductor.
Background reading. Rev. Mod. Phys. 71, S358 (scaling, universality and renormalization as the three pillars of critical phenomena).	

— Cluster B — The renormalization group at work

Paper	Comment
Phys. Rev. Lett. 28, 240	The ε -expansion in four pages; a paper one should be able to reproduce on a blackboard.
Phys. Rev. B 4, 3174	The recursion formula that made Kadanoff’s block-spin picture quantitative.
Phys. Rev. Lett. 31, 1411	The first real-space renormalization-group calculation, on the triangular lattice: position-space RG as a research paper.
Phys. Rev. Lett. 36, 691	The other ε -expansion, coming up from the lower critical dimension in $2 + \varepsilon$ via the nonlinear sigma model.
Phys. Rev. Lett. 39, 95	Borel-resummed perturbation series producing critical exponents that rival experiment: field theory learns to predict numbers.
arXiv:1004.4486	Five-digit exponents of the three-dimensional Ising class from Monte Carlo finite-size scaling: the lattice’s answer to the same question.
Phys. Rev. B 68, 174518	The most precise critical exponent ever measured, in orbit. The talk must also cover its fate: the 2020 bootstrap determination (arXiv:1912.03324) disagrees with it, and theory won.
Phys. Rev. Lett. 29, 917	Interactions decaying as a power law give $\eta = 2 - \sigma$ exactly. The talk must also cover the fate of its crossover claim, corrected in Phys. Rev. B 8, 281 .
Phys. Rev. Lett. 40, 1610	A critical point hiding in the zeros of the partition function, governed by a cubic field theory with ε counted down from six dimensions.

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Paper	Comment (continued)
J. Phys. C 6, 1181	A phase transition without an order parameter: vortex unbinding. (Kosterlitz, J. Phys. C 7, 1046 (1974), is background for the renormalization-group flow.)
Phys. Rev. Lett. 39, 1201	The renormalization group predicts one universal number for the superfluid density at the transition, and experiment can check it.
Phys. Rev. B 14, 1165	Imaginary time as an extra dimension: the Euclidean path integral turned into quantum phase transitions in $d + z$ dimensions.
Background reading. Rev. Mod. Phys. 70, 653 (the basis and formulation of renormalization-group theory) and Rev. Mod. Phys. 55, 583 (Wilson's Nobel lecture).	

— Cluster C — Scale versus conformal invariance

Paper	Comment
JETP Lett. 12, 381	Three pages conjecturing that critical fluctuations are not only scale but conformally invariant, and what that fixes. The founding document of the second half of the course.
Nucl. Phys. B 303, 226	The virial current, and the proof that in two dimensions scale invariance implies conformal invariance.
arXiv:hep-th/0504197	The theory of elasticity is scale invariant but not conformal: a counterexample every engineer already knows.
arXiv:1101.5385	Away from four dimensions, free Maxwell theory is unitary and scale invariant but not conformal, and conformality can only be restored at the price of unitarity.
arXiv:1106.2540	A claimed interacting theory with scale but not conformal invariance, built on a renormalization-group limit cycle. The talk must also cover its fate: arXiv:1204.5221 and the authors' own arXiv:1208.3674 showed that the cycles are conformal in disguise.
arXiv:1501.01776	A non-perturbative renormalization-group argument that scale invariance implies conformal invariance for the three-dimensional Ising model. The argument was challenged in arXiv:1802.02319 and defended in arXiv:1802.07157 ; the talk should know where the dispute stands.
arXiv:1503.02011	The same question put to a Monte Carlo simulation: map the plane to a sphere and see whether the lattice agrees.
Background reading. arXiv:1302.0884 (scale invariance versus conformal invariance, a review).	

— Cluster D — What conformal symmetry buys

Paper	Comment
JETP Lett. 43, 730	The c -theorem: renormalization-group flow in two dimensions is irreversible, and something is lost monotonically. Three pages.
Phys. Lett. B 215, 749	The conjecture that the Euler anomaly coefficient is the four-dimensional analogue of c . The talk must also cover its fate: proved twenty-three years later in arXiv:1107.3987 .
J. Phys. A 17, L385	Map the plane to a cylinder and the correlation length on a strip becomes a universal number fixed by a scaling dimension.
Phys. Rev. Lett. 56, 742 and Phys. Rev. Lett. 56, 746	Assigned as a pair: the finite-size free energy measures the central charge. Transfer-matrix physics a statistical-mechanics audience already owns.

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Paper	Comment (continued)
Phys. Rev. Lett. 52, 1575	Unitarity plus conformal symmetry allows only a discrete series of central charges below one, hence a discrete list of possible two-dimensional critical exponents.
Sov. J. Nucl. Phys. 44, 529	The discrete series of the previous entry identified with Landau–Ginzburg multicritical points: the bridge from Landau theory to conformal field theory.
Phys. Rev. Lett. 54, 1354	The Yang–Lee edge singularity is a non-unitary conformal field theory with $c = -22/5$: what unitarity was protecting us from.
arXiv:hep-th/9111026	A conformal-field-theory calculation predicts a percolation crossing probability that can be measured by counting clusters.
arXiv:quant-ph/0211074	The entanglement entropy of a critical chain grows as $(c/3) \log L$: the central charge appears in a quantity a quantum-mechanics audience understands. (arXiv:hep-th/0405152 is background only.)
Background reading. arXiv:hep-th/9108028 (Ginsparg's Les Houches lectures on applied conformal field theory) and J. Cardy, <i>Scaling and Renormalization in Statistical Physics</i> (Cambridge University Press, 1996).	

— Cluster E — Scale invariance in the wild

Paper	Comment
J. Phys. C 3, 2436	The renormalization group done by hand on a magnetic impurity: a running coupling and an emergent scale.
Phys. Rev. Lett. 56, 889	Scaling and universality for a growing surface, far from equilibrium, with an exact exponent in one dimension.
Phys. Rev. Lett. 64, 1361	Anomalous dimensions in a classical nonlinear diffusion equation: the renormalization group with no statistical mechanics in sight.
Phys. Rev. A 55, R853	A hidden $SO(2, 1)$ symmetry makes a trapped two-dimensional gas breathe at exactly twice the trap frequency.
arXiv:0706.3746	The Schrödinger group as a conformal symmetry; an operator–state correspondence gives the energies of trapped unitary fermions.
Phys. Rev. Lett. 105, 095302	Classical scale invariance of a two-dimensional gas is broken by quantization, visibly, as a shift of a breathing frequency. (Phys. Rev. Lett. 108, 185303 and the experiments Phys. Rev. Lett. 121, 120401 and Phys. Rev. Lett. 121, 120402 are background.)
arXiv:hep-ph/0703260	What a scale-invariant sector would look like at a collider: a non-integer number of particles.
arXiv:0801.1140	How unitarity bounds on scaling dimensions constrain the previous entry.
Phys. Rev. Lett. 30, 1343 and Phys. Rev. Lett. 30, 1346	Assigned as a pair: asymptotic freedom, that is, a renormalization-group flow to a free ultraviolet fixed point, in six pages total.
Nucl. Phys. B 196, 189	The opposite flow: a weakly coupled infrared fixed point in a gauge theory, exactly as in Wilson–Fisher.
Background reading. arXiv:1101.4886 (a tutorial on scale and conformal symmetries in diverse dimensions).	

— Cluster F — The conformal bootstrap

Paper	Comment
Sov. Phys. JETP 39, 10	Replace the Lagrangian by crossing symmetry and unitarity, and solve: the original bootstrap, two decades ahead of its tools.
arXiv:0905.2211	How a rigorous bound comes out of a crossing equation; the shortest route into the numerical bootstrap.
arXiv:1307.3111	A bootstrap without positivity: truncate the spectrum and solve the crossing equations directly.
arXiv:1309.5089	The bootstrap kink followed from $d = 4$ down to $d = 2$, connecting the ε -expansion to Onsager.
arXiv:1512.00013	In non-integer dimension the Wilson–Fisher theory has negative-norm states: unitarity and the ε -expansion do not mix.
arXiv:1406.4858	From a kink on a boundary to an island. The numerical appendices are skippable; the conceptual body is a dozen pages.
arXiv:1505.00963	Wilson–Fisher anomalous dimensions with no Feynman diagram: multiplet recombination plus the equation of motion.
arXiv:1602.07295	The bootstrap as a falsifier: necessary conditions for a discrete symmetry to enhance to a continuous one in the infrared, and what they rule out.
Background reading. arXiv:1805.04405 (the review in Rev. Mod. Phys.), arXiv:1601.05000 (Rychkov's EPFL lectures) and arXiv:1602.07982 (Simmons-Duffin's TASI lectures).	